031103

272. da Silva C L, Sonnenfeld R G, Edens H E, Krehbiel P R, Quick M G, and Koshak W J 2019 The Plasma Nature of Lightning Channels and the Resulting Nonlinear Resistance *Journal of Geophysical Research: Atmospheres* **124** 9442-9463
273. Marinov I, Starikovskaia S, and Rousseau A 2014 Dynamics of plasma evolution in a nanosecond underwater discharge *Journal of Physics D: Applied Physics* **47** 224017
274. Fujita H, Kanazawa S, Ohtani K, Komiya A, and Sato T 2013 Spatiotemporal analysis of propagation mechanism of positive primary streamer in water *J. Appl. Phys.* **113** 113304
275. Luque A and Ebert U 2014 Growing discharge trees with self-consistent charge transport: the collective dynamics of streamers *New J. Phys.* **16** 013039
276. Niemeyer L, Pietronero L, and Wiesmann H J 1984 Fractal Dimension of Dielectric-Breakdown *Phys. Rev. Lett.* **52** 1033-1036
277. Luque A and Ebert U 2011 Electron density fluctuations accelerate the branching of positive streamer discharges in air *Phys. Rev. E* **84** 046411
278. Arrayás M, Betelú S, Fontelos M A, and Trueba J L 2008 Fingering from Ionization Fronts in Plasmas *SIAM Journal on Applied Mathematics* **68** 1122-1145
279. Dubietis A, Tamošauskas G, Šuminas R, Jukna V, and Couairon A 2017 Ultrafast supercontinuum generation in bulk condensed media *Lithuanian Journal of Physics* **57** 113–157
280. Bulgakova N M, Zhukov V P, and Meshcheryakov Y P 2013 Theoretical treatments of ultrashort pulse laser processing of transparent materials: toward understanding the volume nanograting formation and “quill” writing effect *Appl. Phys. B.* **113** 437-449
281. Rudenko A, Colombier J-P, Höhm S, Rosenfeld A, Krüger J, Bonse J, and Itina T E 2017 Spontaneous periodic ordering on the surface and in the bulk of dielectrics irradiated by ultrafast laser: a shared electromagnetic origin *Sci. Rep.* **7** 12306
282. Bonse J and Gräf S 2020 Maxwell Meets Marangoni - A Review of Theories on Laser‐Induced Periodic Surface Structures *Laser & Photonics Reviews* **14** 2000215
283. Bonse J and Sokolowski‐Tinten K 2024 Probing Laser‐Driven Structure Formation at Extreme Scales in Space and Time *Laser & Photonics Reviews* **18** 2300912
284. Warren S G 1984 Optical-Constants of Ice from the Ultraviolet to the Microwave *Appl. Opt.* **23** 1206-1225
285. Prahl S, "Optical Absorption of water, available at http://omlc.ogi.edu/spectra/water/index.html", retrieved 09.02.2024.
286. Henke B L, Gullikson E M, and Davis J C 1993 X-ray interactions - photoabsorption, scattering, transmission, and reflection at E=50-30,000 Ev, Z=1-92 *Atom. Data. Nucl. Data.* **54** 181-342
287. Henke B L, "Optical constants, available at https://henke.lbl.gov/optical_constants/atten2.html", retrieved 09.02.2024.
288. Meesungnoen J, Jay-Gerin J P, Filali-Mouhim A, and Mankhetkorn S 2002 Low-energy electron penetration range in liquid water *Radiat. Res.* **158** 657-660
289. Bulgakova N M, et al. 2016 Ultrashort-pulse laser processing of transparent materials: insight from numerical and semi-analytical models *Proc. SPIE* **9735** 97350N
290. Petrov G M, Palastro J P, and Peñano J 2017 Bremsstrahlung from the interaction of short laser pulses with dielectrics *Phys. Rev. E* **95** 053209

Accepted Manuscript

291. Ghisellini G 2013 *Bremsstrahlung and Black Body in Radiative Processes in High Energy Astrophysics* (Springer) Vol. 873 pp. 23-29

292. Aghaei M, Mehrabian S, and Tavassoli S H 2008 Simulation of nanosecond pulsed laser ablation of copper samples: A focus on laser induced plasma radiation *J. Appl. Phys.* **104** 053303

293. Alcock A J, DeMichelis C, Hamal K, and Tozer B A 1968 Expansion Mechanism in a Laser-Produced Spark *Phys. Rev. Lett.* **20** 1095-1097

294. Bekefi G 1966 *Radiation processes in plasmas* Wiley series in Plasma Physics (Wiley)

295. Camara C, Putterman S, and Kirilov E 2004 Sonoluminescence from a single bubble driven at 1 megahertz *Phys. Rev. Lett.* **92** 124301

296. Roberts R M, Cook J A, Rogers R L, Gleeson A M, and Griffy T A 1996 The energy partition of underwater sparks *J. Acoust. Soc. Am.* **99** 3465-3475

297. Carr C W, Radousky H B, Rubenchik A M, Feit M D, and Demos S G 2004 Localized Dynamics during Laser-Induced Damage in Optical Materials *Phys. Rev. Lett.* **92** 087401

298. Hammer D and Frommhold L 2001 Topical review - Sonoluminescence: how bubbles glow *J Mod Optic* **48** 239-277

299. Aguilera J A and Aragón C 2007 Multi-element Saha–Boltzmann and Boltzmann plots in laser-induced plasmas *Spectrochimica Acta Part B: Atomic Spectroscopy* **62** 378-385

300. Arons A B and Yennie D R 1948 Energy Partition in Underwater Explosion Phenomena *Rev Mod Phys* **20** 519-583

301. Kumar S, Maxwell I Z, Heisterkamp A, Polte T R, Lele T P, Salanga M, Mazur E, and Ingber D E 2006 Viscoelastic Retraction of Single Living Stress Fibers and Its Impact on Cell Shape, Cytoskeletal Organization, and Extracellular Matrix Mechanics *Biophys. J.* **90** 3762-3773

302. Supatto W, Debarre D, Moulia B, Brouzes E, Martin J L, Farge E, and Beaurepaire E 2005 In vivo modulation of morphogenetic movements in Drosophila embryos with femtosecond laser pulses *Proc. Natl. Acad. Sci. U.S.A.* **102** 1047-1052

303. Segawa Y, Minoshima W, Masui K, and Hosokawa C 2024 Neuronal Electrical Activity in Neuronal Networks Induced by a Focused Femtosecond Laser *ACS Omega* **10** 1354-1363

304. Ma S, Dong B, Clark M G, Everly R M, Mahapatra S, and Zhang C 2025 Real‐Time and Site‐Specific Perturbation of Dynamic Subcellular Compartments Using Femtosecond Pulses *Small Science* 2500166

305. Krasieva T, Chapman C, Morte V L, Venugopalan V, and Tromberg B 1998 Mechanisms of cell permeabilization by laser microirradiation *Proc. SPIE* **3260** 38-44

306. Vogel A, Linz N, Freidank S, and Liang X 2010 Controlled nonlinear energy deposition in transparent materials: Experiments and theory *AIP Conf. Proc.* **1278** 51-55

307. Vogel A, Linz N, and Freidank S 2013 Method for laser machining transparent materials *US Patent* **8350183**

308. Coohill T P 2002 *Uses and effects of ultraviolet radiation on cells and tissues* in Lasers in Medicine 1 ed. edited by Waynant R W (CRC Press)

309. Freidank S, Vogel A, and Linz N 2020 Optical Vortex Beam for Gentle and Ultraprecise Intrastromal Corneal Dissection in Refractive Surgery *Transl Vis Sci Techn* **9** 22

310. Habbema L, Verhagen R, Van Hal R, Liu Y, and Varghese B 2012 Minimally invasive non-thermal laser technology using laser-induced optical breakdown for skin rejuvenation *J. Biophotonics* **5** 194-199

311. Tanghetti E A 2016 The histology of skin treated with a picosecond alexandrite laser and a fractional lens

Accepted Manuscript

array *Laser Surg. Med.* **48** 646-652

312. Yildirim M, Ferhanoglu O, Kobler J, Zeitels S M, and Ben-Yakar A 2013 Parameters affecting ultrafast laser microsurgery of subepithelial voids for scar treatment in vocal folds *J. Biomed. Opt.* **18** 118001
313. Lieberman S, et al. 2024 Circumscribing Laser Cuts Attenuate Seizure Propagation in a Mouse Model of Focal Epilepsy *Advanced Science* **11** 2300747
314. Schaffer C B, Garcia J F, and Mazur E 2003 Bulk heating of transparent materials using a high-repetition-rate femtosecond laser *Appl. Phys. A* **76** 351-354
315. Eaton S M, Zhang H, Ng M L, Li J Z, Chen W J, Ho S, and Herman P R 2008 Transition from thermal diffusion to heat accumulation in high repetition rate femtosecond laser writing of buried optical waveguides *Opt. Express* **16** 9443-9458
316. Riahi F, Doñate-Buendia C, Barcikowski S, and Gökce B 2023 How Nanoparticle Size and Bubble Merging Is Governed by Short‐Range Spatially Controlled Double‐Beam Laser Ablation in Liquids *Particle & Particle Systems Characterization* **41** 2300145
317. Apitz I and Vogel A 2005 Material ejection in nanosecond Er:YAG laser ablation of water, liver, and skin *Appl. Phys. A* **81** 329-338
318. Bechtel J H, Lee Smith W, and Bloembergen N 1977 Two-photon photoemission from metals induced by picosecond laser pulses *Phys. Rev. B* **15** 4557-4563
319. Stoian R, Rosenfeld A, Ashkenasi D, Hertel I V, Bulgakova N M, and Campbell E E B 2002 Surface Charging and Impulsive Ion Ejection during Ultrashort Pulsed Laser Ablation *Phys. Rev. Lett.* **88** 097603
320. Bulgakova N M, Stoian R, Rosenfeld A, Hertel I V, and Campbell E E B 2004 Electronic transport and consequences for material removal in ultrafast pulsed laser ablation of materials *Phys. Rev. B* **69** 054102
321. Amoruso S 1999 Modeling of UV pulsed-laser ablation of metallic targets *Appl. Phys. A* **69** 323-332
322. Bulgakova N M and Bulgakov A V 2001 Pulsed laser ablation of solids: transition from normal vaporization to phase explosion *Applied Physics A Materials Science & Processing* **73** 199-208
323. Winter J, Redka D, Minár J, Schmidt M, and Huber H P 2023 Resolving transient temperature and density during ultrafast laser ablation of aluminum *Appl. Phys. A* **129** 665
324. Wei H, Li Q, Nie C, Lu J, Tan X, Li J, and He W 2025 Influence of electron thermal capacity on properties of pulsed laser ablation on Au in high energy domain *Phys. Lett. A* **529** 130094
325. Legall H, Schwanke C, Bonse J, and Krüger J 2019 The influence of processing parameters on X-ray emission during ultra-short pulse laser machining *Appl. Phys. A* **125** 570
326. Sakka T, Masai S, Fukami K, and Ogata Y H 2009 Spectral profile of atomic emission lines and effects of pulse duration on laser ablation in liquid *Spectrochimica Acta Part B: Atomic Spectroscopy* **64** 981-985
327. Tamura A, Matsumoto A, Fukami K, Nishi N, and Sakka T 2015 Simultaneous observation of nascent plasma and bubble induced by laser ablation in water with various pulse durations *J. Appl. Phys.* **117** 173304
328. Tomko J, Naddeo J J, Jimenez R, Tan Y, Steiner M, Fitz-Gerald J M, Bubb D M, and O'Malley S M 2015 Size and polydispersity trends found in gold nanoparticles synthesized by laser ablation in liquids *Phys. Chem. Chem. Phys.* **17** 16327-16333
329. Mehta K and Baruah P K 2022 A comprehensive review and outlook on the experimental techniques to investigate the complex dynamics of pulsed laser ablation in liquid for nanoparticle synthesis *Rev. Sci. Instrum.* **93** 091501
330. Song X, Wu X Q, Xiao K L, Li C, Wang H Y, and Jiang M Q 2020 Nanosecond laser ablation of a metallic glass

Accepted Manuscript